

Automation System Design

Smart Home Lighting System

CodeAlpha Internship — Robotics & Automation

Task 3: Automation System Design

1. Objective

The goal of this design is to automate residential lighting so that lights switch on only when a room is occupied and ambient light is insufficient, and switch off automatically once the room is vacated. This removes the need for manual switching, reduces electricity wastage from lights left on in empty or well-lit rooms, and adds a basic layer of convenience and energy efficiency to a home.

2. System Components

The system is built from a small set of low-cost components, summarized below.

Sensor inputs

PIR motion sensor, LDR (light-dependent resistor)

Controller

Microcontroller (e.g. Arduino Uno / ESP32)

Actuator

Relay module switching the light circuit

Timer window

5 minutes of no motion before shutoff

Trigger condition

Motion detected AND ambient light below threshold

- PIR motion sensor: detects the presence and movement of a person within its field of view using infrared radiation.
- LDR sensor: measures ambient light intensity, allowing the system to distinguish daylight from darkness.
- Micro controller: continuously reads both sensors and runs the control logic.
- Relay module: acts as an electronic switch that the micro controller uses to turn the AC light circuit on or off.

3. System Logic

The control logic follows a continuous monitoring loop, illustrated in the accompanying flowchart. At every cycle, the controller checks the PIR sensor for motion. If no motion is detected, the system simply continues monitoring. If motion is detected, the controller then checks the LDR reading against a preset brightness threshold. If the room is already bright enough, the lights are left off and the system returns to monitoring — this avoids switching on lights unnecessarily during daytime. If motion is present and the room is dark, the controller turns the lights on and starts a countdown timer, typically set to five minutes. While the timer runs, the system keeps checking the PIR sensor. Any new motion resets the timer, keeping the lights on for as long as the room stays occupied. If the timer reaches zero with no further motion detected, the controller switches the lights off and the system returns to its initial monitoring state, ready to

repeat the cycle.

4. Flowchart

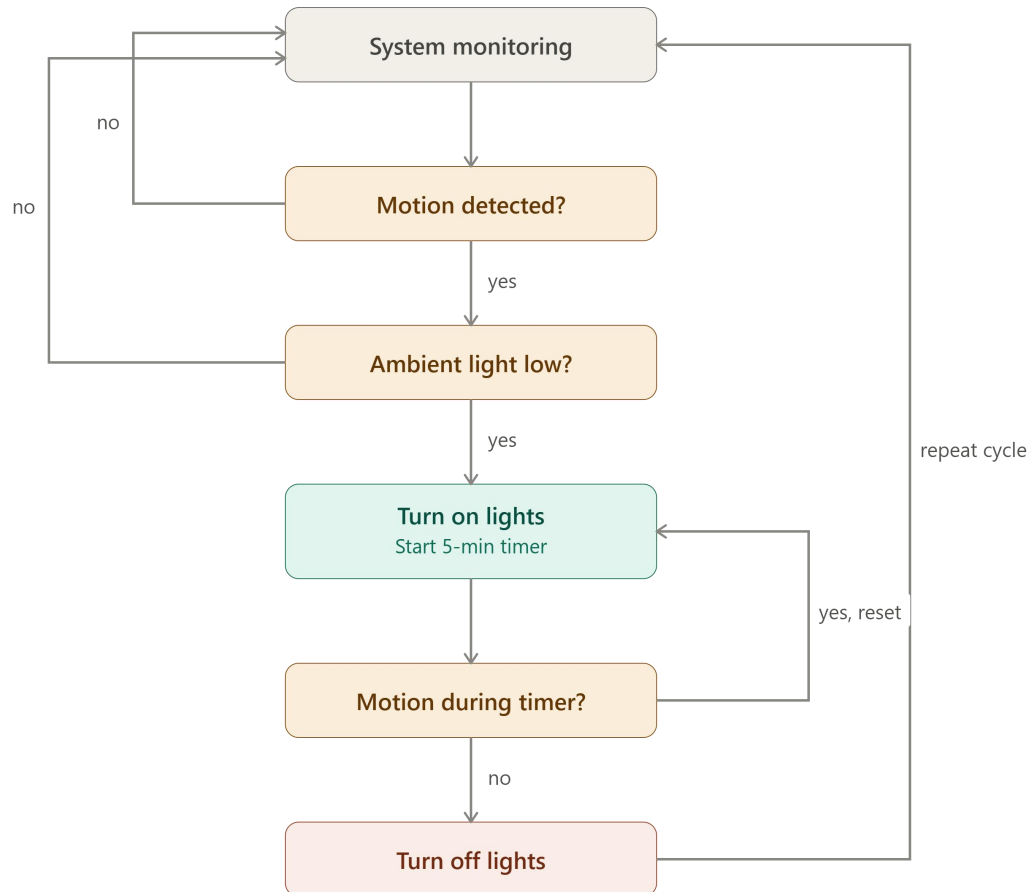


Figure 1: Flowchart of Smart Home Lighting System

5. Advantages

- Energy savings: lights are never left on in an empty or already-bright room.
- Convenience: no manual switches needed; the system responds automatically to occupancy.
- Scalability: the same sensor-decision-actuator pattern extends to other automated systems such as conveyor belts, HVAC control, or security lighting.
- Low cost: built entirely from inexpensive, widely available components.

6. Possible Extensions

- Add a real-time clock so the system only arms itself after dusk.
- Add Wi-Fi connectivity for remote monitoring and manual override via a mobile app.
- Use dilemma LED drivers instead of a simple relay for gradual brightness control.

